Academy Lab Projects – Showcase

Natalie Koa Hao Yee, 2510016
Lester Liau Jiun Lang, 2509986

XYZ University
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Business scenario overview

- Problem Statement
  - XYZ University's admissions web application experiences slow performances and downtimes during peak enrollment period due to high traffic volume which impacts the users experiences and enrollment efficiency negatively.

- Solution requirement
  - **Functional:** Full CRUD operations (view, add, edit, delete) with no perceivable delay.
  - **Load Balanced:** Distribute traffic across multiple web servers.
  - **Scalable:** Auto-scale to meet variable demand.
  - **Highly Available:** Minimize downtime with multi-AZ deployment.
  - **Secure:** Private database, restricted ports, credentials in AWS Secrets Manager.
  - **Cost Optimized:** Right-sized resources with budget awareness.
  - **High Performing:** Responsive under normal, variable, and peak loads

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Solution overview

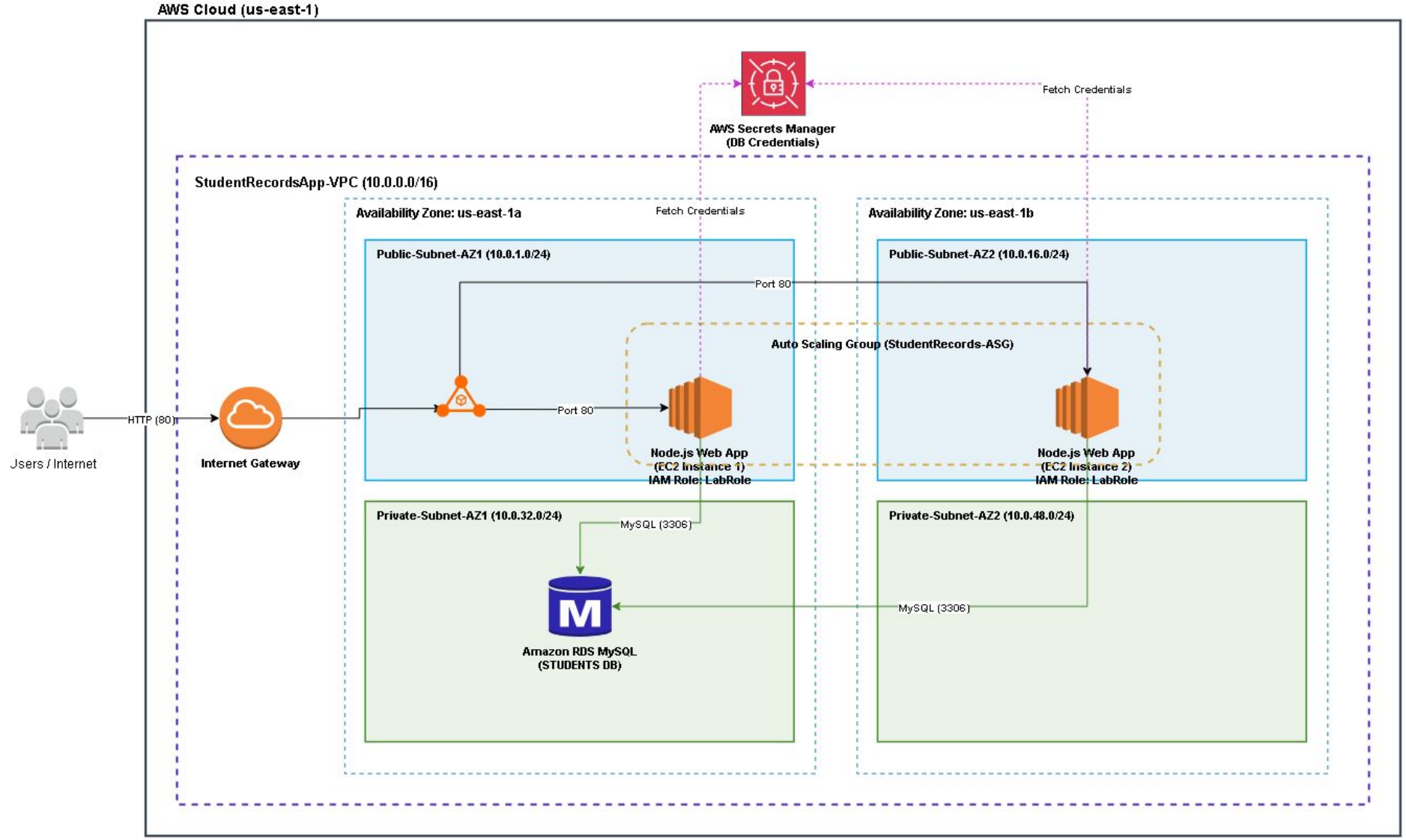
- A cloud-native architecture on AWS hosting the student records application. Uses Application Load Balancer to distribute traffic across EC2 instances managed by Auto Scaling, with Amazon RDS database in private subnets across two Availability Zones.

- **Design Consideration:**
  - **Multi-AZ Deployment:** Web servers span two Availability Zones for fault tolerance
  - **Auto Scaling:** Dynamic scaling based on CPU utilization threshold (70%)
  - **Security:** Least-privilege security groups; database in private subnet; credentials in AWS Secrets Manager
  - **Cost Optimization:** t3.micro instances; balanced capacity settings

- **Use Cases:**
  - View, add, edit, and delete student records
  - Handle peak admissions traffic through automatic scaling
  - Maintain availability during infrastructure failures

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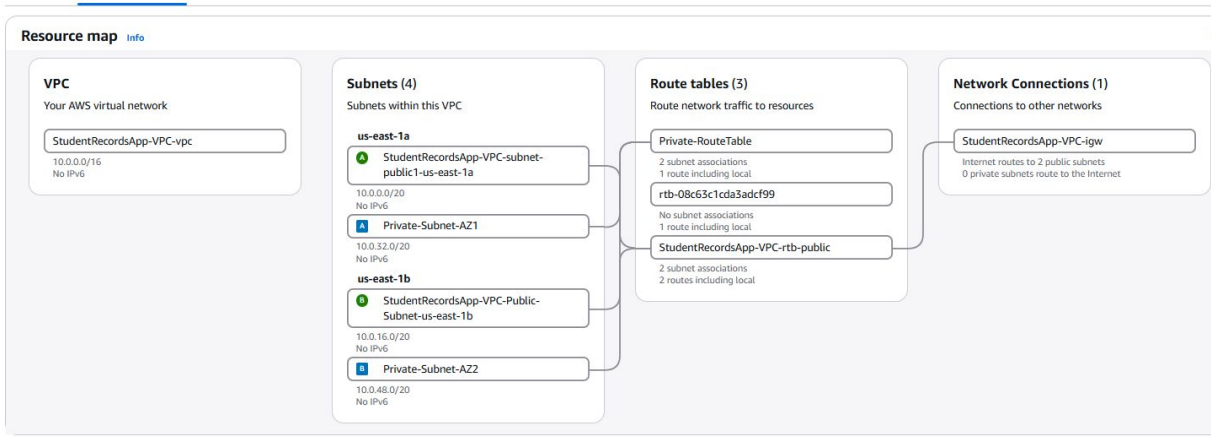
Architecture diagram



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C4 Highly Available - Multi-AZ & VPC Configuration

VPC Resource Map



EC2 Instance Dashboard

☐	StudentRecordsApp-Server-RDS	i-03d19d28acbb693f3	Running	t3.micro	3/3 checks passed	us-east-1a	ec2-3-89-96-152.comp…	3.89.96.152	–	–
☐		i-0abab3bb410a5a263	Running	t3.micro	3/3 checks passed	us-east-1a	ec2-100-58-92-124.co…	100.58.92.124	–	–
☐		i-0005c3ef51049444e	Running	t3.micro	3/3 checks passed	us-east-1b	ec2-54-198-68-40.com…	54.198.68.40	–	–

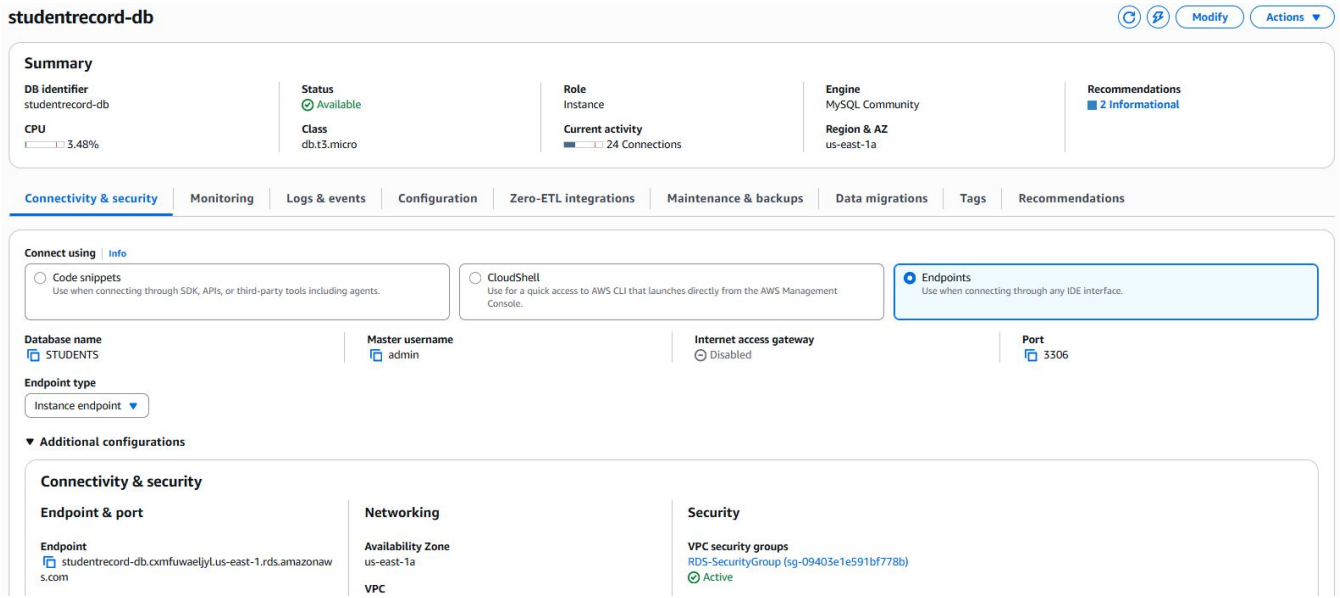
Subnet Table

☐	Name	Subnet ID	State	VPC	Block Public…	IPv4 CIDR	IPv6 CIDR	IPv6 CIDR association
☐	StudentRecordsApp-VPC-subnet-public1-us-east-1a	subnet-0577324da34a76474	Available	vpc-07c25efae5b7e5973 | Stud…	Off	10.0.0.0/20	–	–
☐	StudentRecordsApp-VPC-Public-Subnet-us-east-1b	subnet-0a78fa71468171001	Available	vpc-07c25efae5b7e5973 | Stud…	Off	10.0.16.0/20	–	–
☐	Private-Subnet-AZ2	subnet-0d542eab0dfe035ad	Available	vpc-07c25efae5b7e5973 | Stud…	Off	10.0.48.0/20	–	–
☐	Private-Subnet-AZ1	subnet-0c72009f673126ec7	Available	vpc-07c25efae5b7e5973 | Stud…	Off	10.0.32.0/20	–	–

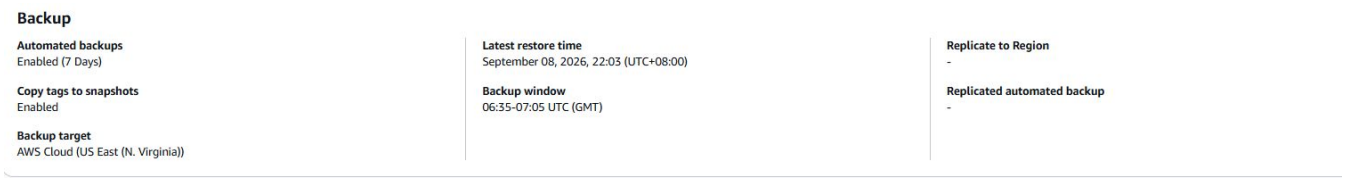
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C4 Highly Available - Database Decoupling & Disaster Recovery Readiness

Decoupled Database



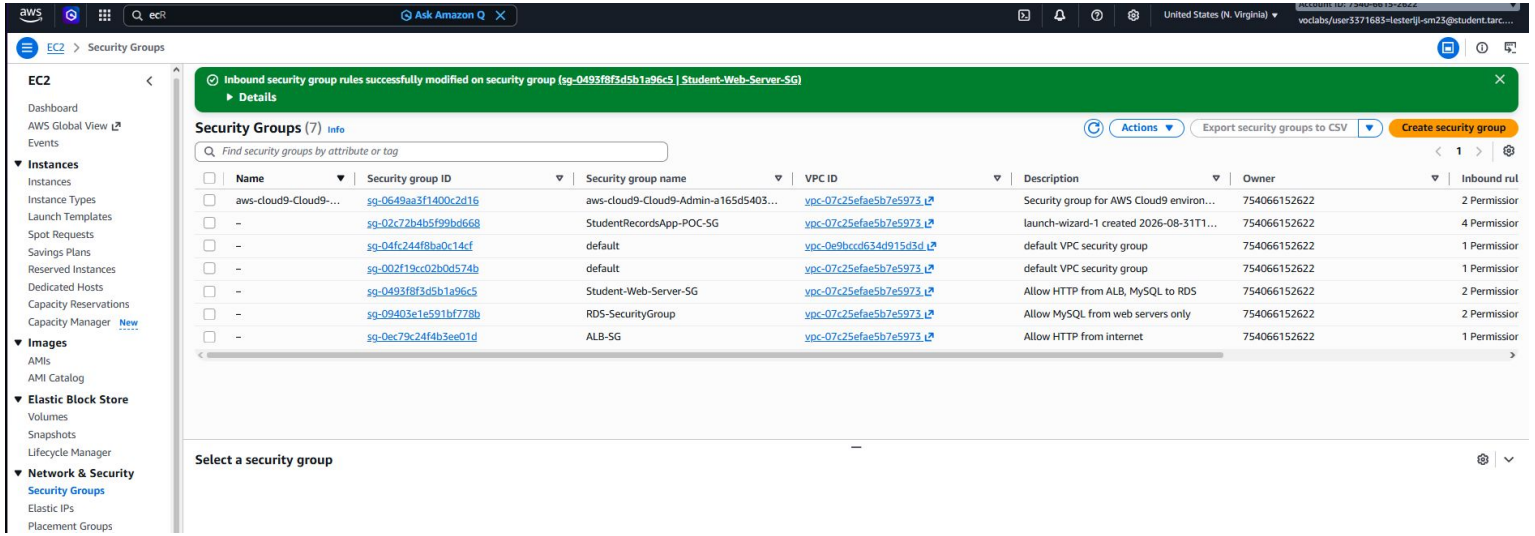
Automated Backup Configuration



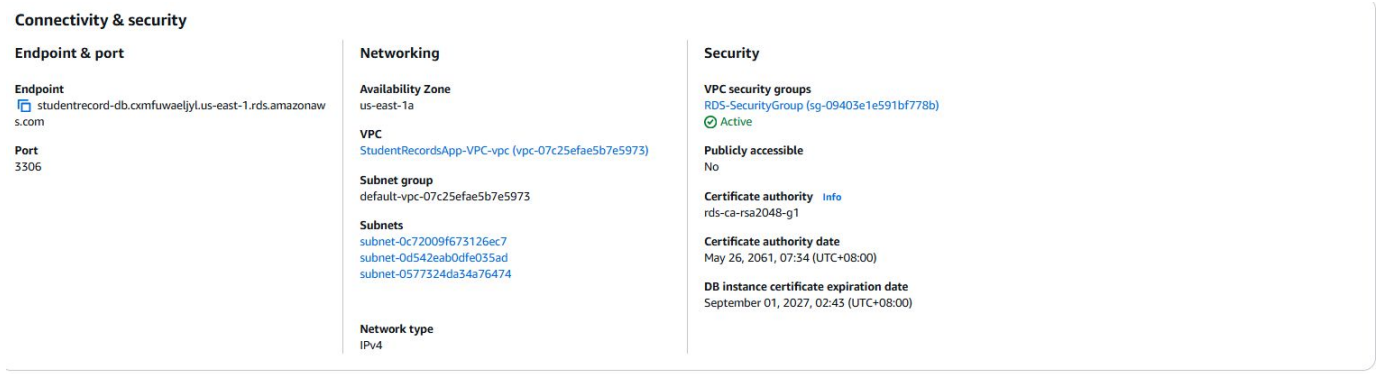
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C5 Security – Network Tiering & Ingress Control

Security Group Table



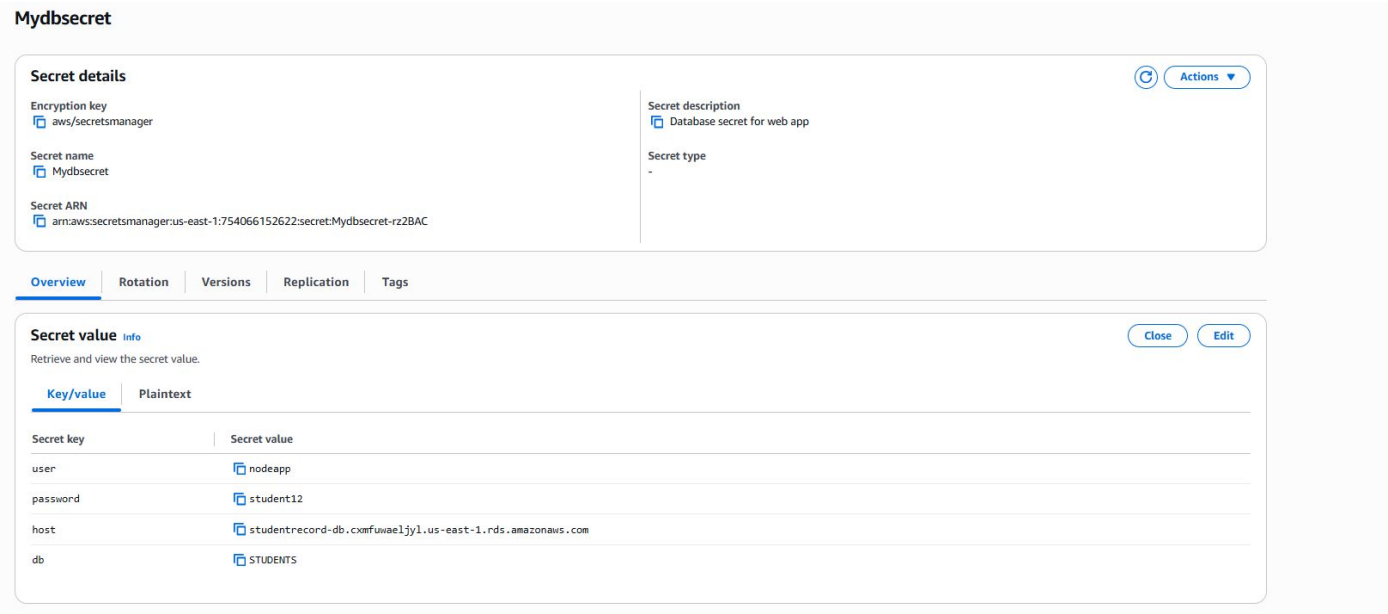
Private Subnet Isolation



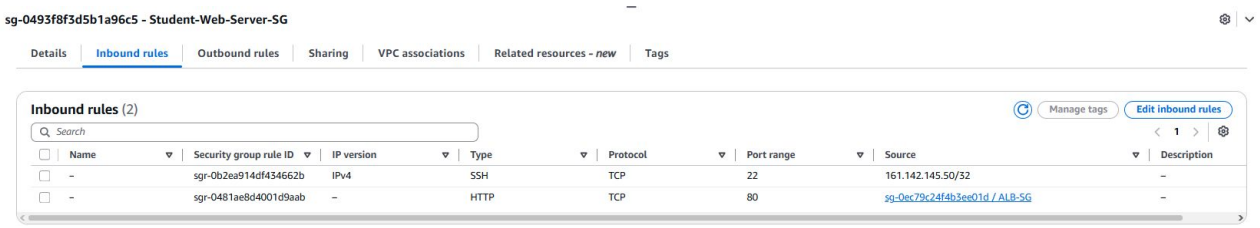
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C5 Security – Credential Externalization & Admin Controls

AWS Secrets Manager Integration



Restricted Administrative Access



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C6 Cost Optimized - 12-Month AWS Calculator Estimate

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Contact your AWS representative: Contact Sales

Export Date: 09/09/2026 **Language:** English

Estimate url

Estimate summary

Upfront cost	Monthly cost	Total 12 months cost
0.00 USD	84.00 USD	1,008.00 USD Includes upfront cost

Detailed Estimate

Name	Group	Region	Upfront cost	Monthly cost
Amazon EC2	-	US East (N. Virginia)	0.00 USD	12.08 USD
Status	-			
Description:	-			
Config summary	Tenancy (Shared Instances), Operating system (Ubuntu Pro), Workload (Consistent, Number of instances: 2), Advance EC2 instance (t3.micro), Pricing strategy (Compute Savings Plans 3yr No Upfront), Enable monitoring (disabled), EBS Storage amount (8 GB), DT Inbound: Not selected (0 TB per month), DT Outbound: Not selected (0 TB per month), DT Intra-Region: (0 TB per month)			

Name	Group	Region	Upfront cost	Monthly cost
Amazon RDS for MySQL	-	US East (N. Virginia)	0.00 USD	54.86 USD
Status	-			
Description:	-			
Config summary	Storage amount (20 GB), Storage for each RDS instance (General Purpose SSD (gp2)), Nodes (1), Instance type (db.t3.micro), Utilization (On-Demand only) (100 %Utilized/Month), Deployment option (Single-AZ), Pricing strategy (OnDemand)			

Name	Group	Region	Upfront cost	Monthly cost
Elastic Load Balancing	-	US East (N. Virginia)	0.00 USD	16.66 USD
Status	-			
Description:	-			
Config summary	Number of Application Load Balancers (1)			

Name	Group	Region	Upfront cost	Monthly cost
AWS Secrets Manager	-	US East (N. Virginia)	0.00 USD	0.40 USD
Status	-			
Description:	-			
Config summary	Number of secrets (1), Average duration of each secret (30 days), Number of API calls (1 per month)			

Acknowledgement

AWS Pricing Calculator provides only an estimate of your AWS fees and doesn't include any taxes that might apply. Your actual fees depend on a variety of factors, including your actual usage of AWS services. Learn more

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